Maincraft Technology

AIML Internship Task 1: House Price Prediction using Linear Regression

1. Introduction

Machine Learning is widely used for predicting future outcomes based on historical data. In this project, a Linear Regression model was developed to predict California housing prices using various housing-related features from the California Housing dataset.

The objective of this project was to understand the complete machine learning workflow including data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and visualization.

The project was implemented using Python and Scikit-learn in Jupyter Notebook.

2. Dataset Information

The California Housing dataset was used for this project. The dataset is available in the Scikit-learn library and contains information collected from California districts.

The dataset includes several important features such as:

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

The target variable in the dataset is:

- Median House Value (MedHouseVal)

The dataset contains thousands of housing records which makes it suitable for regression analysis and prediction tasks.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to better understand the dataset and identify relationships between different features.

The following steps were performed during EDA:

- Checking dataset shape and structure
- Viewing first few rows of data
- Checking missing values
- Generating statistical summaries
- Creating correlation heatmap

A correlation heatmap was generated using Seaborn to visualize relationships between features and the target variable. It was observed that median income had a strong positive correlation with house prices.

No missing values were found in the dataset, therefore additional data cleaning was not required.

4. Data Preprocessing

The dataset was divided into:

- Features (X)
- Target Variable (y)

The target variable was MedHouseVal.

The dataset was then split into training and testing datasets using the train_test_split() function.

- 80% data was used for training
- 20% data was used for testing

This helped evaluate the model performance on unseen data.

5. Model Training

A Linear Regression model from Scikit-learn was used for training.

Linear Regression is a supervised machine learning algorithm used for predicting continuous numerical values.

The model was trained using the training dataset with the following steps:

1. Creating Linear Regression object

2. Fitting the model on training data
3. Making predictions on test data

The model learned relationships between housing features and house prices.

6. Model Evaluation

The model performance was evaluated using the following metrics:

Mean Absolute Error (MAE)

MAE measures the average prediction error.

Root Mean Squared Error (RMSE)

RMSE measures the standard deviation of prediction errors.

R^2 Score

R^2 Score measures how well the model explains the variance in the data.

Obtained Results

- MAE ≈ 0.53
- RMSE ≈ 0.74
- R^2 Score ≈ 0.57

The obtained R^2 score indicates that the model achieved satisfactory prediction performance for this dataset.

7. Visualization

Several visualizations were created to better understand model performance:

Correlation Heatmap

Used to analyze relationships between variables.

Actual vs Predicted Scatter Plot

Used to compare predicted housing prices with actual values.

Residual Distribution Plot

Used to analyze prediction errors.

These visualizations helped in understanding the strengths and limitations of the Linear Regression model.

8. Conclusion

This project successfully implemented a Linear Regression model on the California Housing dataset.

The project demonstrated the complete machine learning workflow including:

- Data Loading
- Data Exploration
- Data Preprocessing
- Model Training
- Prediction
- Evaluation
- Visualization

The model achieved satisfactory results and was able to predict housing prices with reasonable accuracy.

Future improvements can include:

- Feature Engineering
- Hyperparameter Tuning
- Advanced models such as Random Forest and XGBoost

Overall, this project provided a strong introduction to regression-based machine learning techniques and practical implementation using Python.